Md Fahim Razi Prantik

Junior Researcher
State Key Laboratory of CNS/ATM
School of Electronic and Information Engineering
Beihang University

Email: **fahimraziprantik@gmail.com**
LinkedIn: **Md Fahim Razi Prantik**
ORCID: **0009-0005-2436-4145**
GitHub: **Fahim Razi Prantik**

Education

Beihang University Beijing, China
M.Eng. in Traffic Information Engineering and Control Sep. 2024 – May 2027 (Expected)
Advisor: Prof. Peng Zhao — Defense expected: May 2027.
Thesis: "Joint Optimization of Trajectory, Beamforming, and Power Allocation for UAV-Enabled Integrated Sensing and Communication." Investigates DRL-based joint optimization of UAV trajectory, beamforming, and spectrum management for ISAC systems in urban aerospace environments.
GPA: 3.62 on a scale of 4.00

Civil Aviation University of China Tianjin, China
B.Eng. in Communication Engineering Sep. 2020 – Jul. 2024
GPA: 3.96 on a scale of 4.00 (Average: 93.49%)

Research Interest

UAV Systems, Integrated Sensing and Communication (ISAC), Trajectory Optimization, Air Traffic Management, Deep Reinforcement Learning, Machine Learning.

Research Experience & Projects

Research Experience

Junior Researcher Sep. 2024 – Present
State Key Laboratory of CNS/ATM, Beihang University Beijing, China

- Designing and optimizing UAV-assisted hierarchical ISAC systems for urban environments, addressing trajectory optimization, interference management, and multi-UAV coordination.
- Developing and simulating innovative solutions for next-generation wireless networks and air traffic management systems.

Research Participant Grant No. 2024A03
State Key Laboratory of CNS/ATM, Beijing, China

- Research on Key Technologies of Integrated Sensing and Communication in Airport Surface Based on Aeronautical Broadband Communication System.
- Published: "Energy-Efficient Trajectory Planning and Spectrum Management for AAVs-Based ISAC in Urban Aerospace Applications." AIAA SCITECH 2026 Forum.

Research Participant Grant No. 52302424
National Natural Science Foundation of China

- Participated in research, algorithm design, and implementation for multi-source, multi-domain collaborative reasoning in autonomous air traffic situational awareness.
- Published: "Density-Adaptive UAV ISAC: A DRL Framework for Joint Beam Control and Trajectory Design." IEEE Wireless Communications Letters, 2026.

Academic Projects

ROBOTAC – Humanoid Kung Fu Fighting Robot Apr. 2026
25th CURC ROBOTAC Competition, Beihang Team 1

- Implemented full-contact combat behaviors under competition constraints (2-min match, 1.5 m × 1.5 m arena, size/weight compliance).

Arduino-Based Ultrasonic Radar System Jun. 2023
Mechatronics and PLC Lab, BITAC

- Built an object-detection radar using Arduino Uno, ultrasonic sensor, servo motor, and MATLAB visualisation.

Quad-copter Drone May 2023
Mechatronics and PLC Lab, BITAC

- Designed and built a quad-copter using the KK2.1.5 flight controller; programmed stabilisation and autonomous flight routines.

Innovating Business and Commerce with NLP Apr. 2023
Team Project with Prof. Kris Singh (Tsinghua University)

- Addressed critical questions guiding emerging countries in establishing efficient and intelligent transportation systems using natural language processing.

AI Applications for Transportation System Apr. 2022
Team Project with Prof. Kris Singh (Tsinghua University)

- Addressed critical questions guiding emerging countries in establishing intelligent transportation systems that are more sustainable and efficient.

Publications

Journal Articles

- **Md Fahim Razi Prantik**, Z. Zhang, L. Zhao, K. Cai, P. Zhao. "Density-Adaptive UAV ISAC: A DRL Framework for Joint Beam Control and Trajectory Design." *IEEE Wireless Communications Letters*, 2026. DOI: 10.1109/LWC.2026.3689708
- R. M. Atiqur*, **Md Fahim Razi Prantik***, S. T. Islam, A. Rahman, S. H. Masum, M. A. Hossain. "Hyperdimensional Computing and Ridge Regression for Airfoil Aerodynamic Prediction: A Physics-Informed Machine Learning Framework." *Aerospace Science and Technology*, 2026. DOI: 10.1016/j.ast.2025.111591
  *Equal contribution.

Conference Proceedings

- **Md Fahim Razi Prantik**, Z. Zhang, L. Zhao, K. Cai, Z. Jiao, P. Zhao. "Energy-Efficient Trajectory Planning and Spectrum Management for AAVs-Based ISAC in Urban Aerospace Applications." *AIAA SCITECH 2026 Forum*, 2026. DOI: 10.2514/6.2026-0662
- R. Abdullah, Z. Xi, **Md Fahim Razi Prantik**. "Design of Moving Target Defense and Control Compensation for UAVs under FDI Attacks." *IEEE ICUS*, 2025. DOI: 10.1109/ICUS66297.2025.11295369

Submitted

- **Md Fahim Razi Prantik**, et al. "MARL-Based UAV Trajectory, Energy, and Beamforming Optimization for ISAC System at Urban Intersections." (Submitted)

Leadership & Organizations

- **Deputy Head, Academic & Innovation Department** – International Student Union, Beihang University Sep. 2024 – Present
- **Senior Member** – Beihang International Students Robotics Team Sep. 2024 – Present

Funding & Scholarships

- **Chinese Government Scholarship (CSC)** – Full funding for M.Eng. studies 2024 – 2027
  Awarded by the China Scholarship Council for outstanding international students; covers tuition, accommodation, and stipend for 3 years.
- **First Class Outstanding Enrolled Student Scholarship** – Civil Aviation University of China
  2021, 2022, 2023
  Merit-based scholarship awarded 3 times for top academic performance in Communication Engineering.

Academic Accomplishments

- KT&G Alliance Students' Venture Forum (ASVF), Seoul, South Korea – Presented entrepreneurial ideas on stage, representing Bangladesh Apr. 2026
- Best Design Award (2nd Prize), 3rd Capital Int'l Students Robotics Competition 2026
- Excellent Study Award, First Prize, Beihang University 2025
- Ambassador Award, Second Prize, Beihang University 2025
- 3rd Place, ISRC Control Group – Line Following Challenge 2025
- Outstanding Member Award, ISRC 8 Weeks Program 2025
- Innovation & Entrepreneurship for Digital Economy Award 2023
- Chinese Culture Ambassador & Robotics of the Future (Excellent) 2022

Technical Skills

- **Programming Languages:** Python, C, MATLAB, Arduino, LaTeX
- **AI & Data:** Machine Learning, Deep Learning
- **Aerospace & UAV:** UAV Systems, ISAC, ADS-B Decoding, Trajectory Optimization, Air Traffic Management
- **Languages:** Bengali (Native), English (Professional), Hindi (Speaking), Chinese (HSK 2)

Recommendation

Prof. Yan Xu
Professor
School of Electronic and Information Engineering
Beihang University
Email: yan.xu@buaa.edu.cn
Faculty Page

Prof. Kai Liu
Professor
School of Electronic and Information Engineering
Beihang University
Email: liuk@buaa.edu.cn
Faculty Page